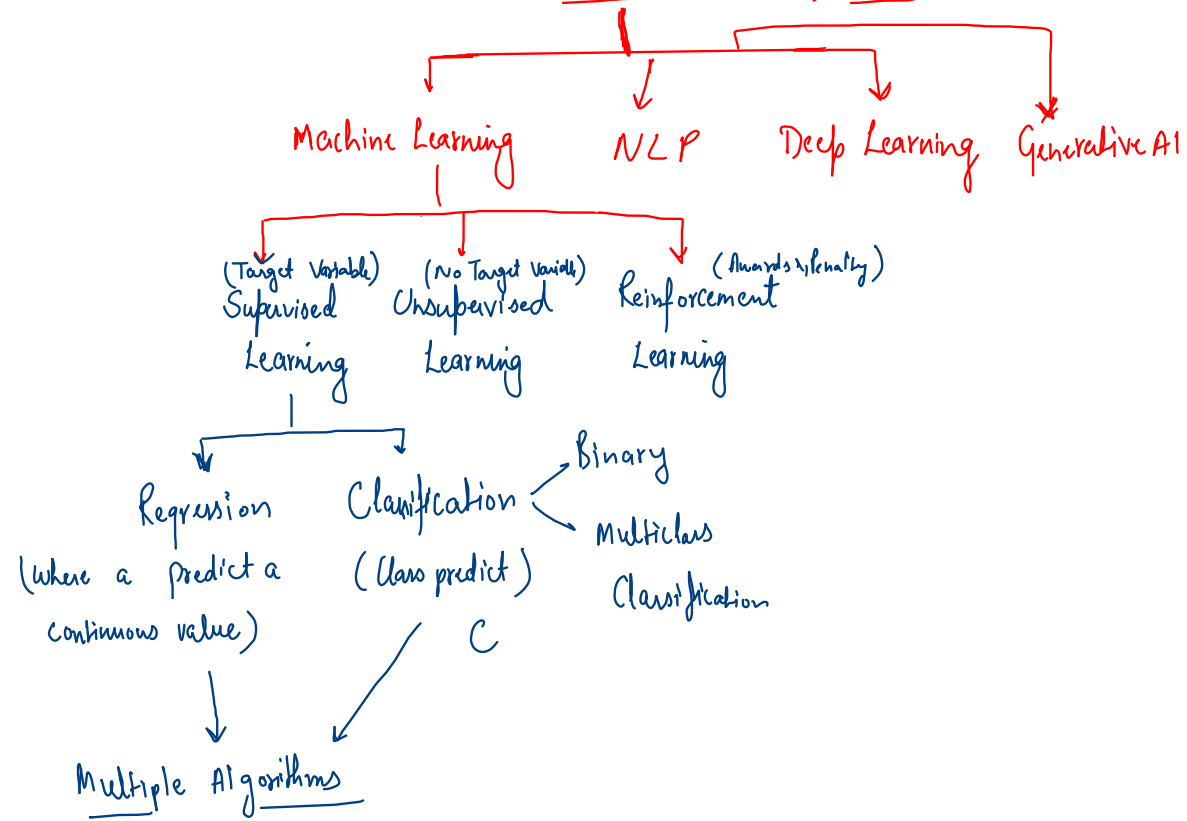
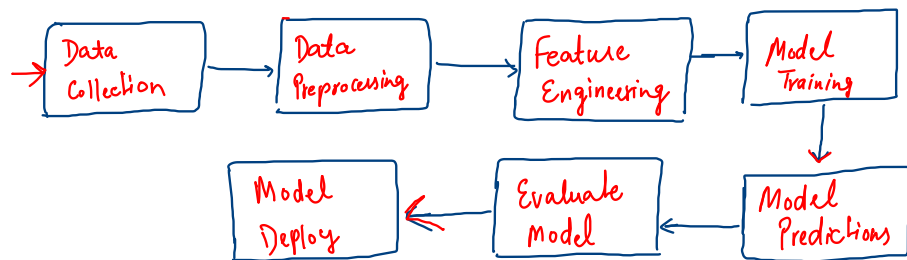


Data Science $\Rightarrow$ AI



Machine Learning Flow



ML Usecase

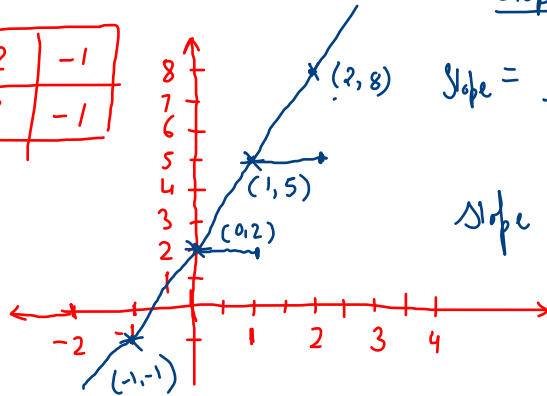
Input / Independent Variables ↑ Feature			Target / Prediction ↓ Dependent Variable
P/E	Revenue	Profit	Stock Price
46	1000000	10000	350
51	200000	45000	323
39	6003450	69000	312
44	7912645	740000	400
⋮	⋮	⋮	⋮

Linear Regression ←

$y = mx + c$ Slope Intercept
 $\Leftrightarrow y = 3x + 2$

x	0	1	2	-1
y	2	5	8	-1

$x_1, y_1 \rightarrow (-1, -1)$
 $x_2, y_2 \rightarrow (2, 8)$



Slope = 3

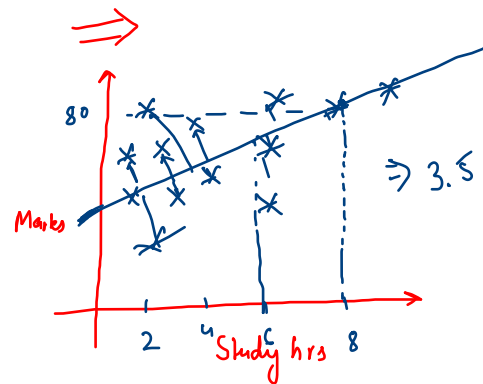
Slope = $\frac{y_2 - y_1}{x_2 - x_1} \Rightarrow$

Slope = $\frac{dy}{dx}$

$y = mx + c$ $\leftarrow$ Single Feature

6

Study hrs	Marks
2	45
4	69
3	74
5	81
1	38



Assumptions

- ① Linearity $\rightarrow$ X and Y should have linear relationship
- ② Homoscedasticity $\rightarrow$ Residuals/Error should have constant variance.
- ③ Normality $\rightarrow$ Residuals should follow distribution
- ④ Independence $\rightarrow$ There should be no correlation b/w residuals